

Pathogen dynamics nested in a host Wright-Fisher model: towards joint host-pathogen genealogies

CoNeMEsSA Seminar

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Outline

- 1 Phylogenetics
- 2 Joint Host-Pathogen Genealogies
- 3 Nested Host and Pathogen model

Phylogenetics: Inferring epidemiological parameters from pathogen genetic sequences.

- Pathogens coalesce backwards in time with the transmission tree of infected individuals.
- We can derive the coalescent rates from an epidemiological model (incidence and prevalence), and compare to observed pathogen sequence trees.

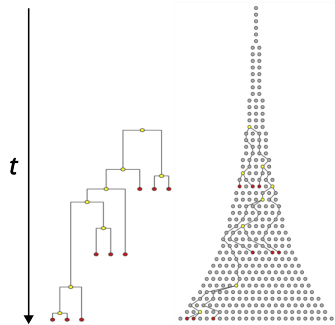


Figure: Infectious branching process from Kühnert et al. 2011.

Phylodynamics: Basics of pathogen coalescence

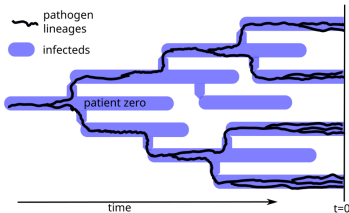


Figure: Pathogen coalescent within transmission tree (J. Müller, 2026)

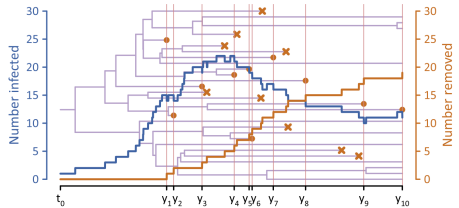


Figure: Coalescent process within the epidemiological dynamics of I (Fig. 1D, Plessis and Stadler 2015)

Phylodynamics: Assumptions

Assumptions

- 1 Host coalescence is negligible compared to pathogen coalescent times.
- 2 Pathogen coalescent tree reflects the transmission tree.
- 3 Each host is infected by a single pathogen, so $I \sim N_{\text{Pathogen}}$
- 4 Evolution is measurable, i.e mutations occur on the timescale of transmission.

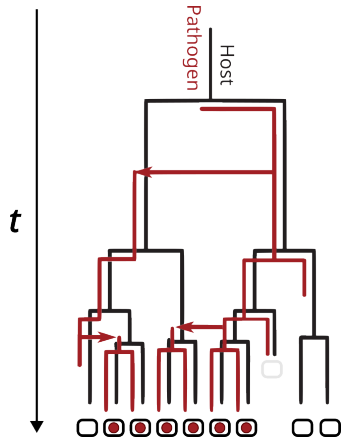
Real-world host-pathogen systems:

- complex disease life-cycles including *varying transmission modes*, vectors and environmental factors
- reciprocal host-pathogen interactions (*coevolution*)

Joint Host and Pathogen Genealogies

Applying phylodynamics to diseases beyond epidemic growth, requires consideration of host coalescence, i.e. **no** time scale separation of host and pathogen coalescence:

The pathogen genealogical process is embedded within the host genealogy, mediated by horizontal and vertical transmission



Vertical and horizontal transmission

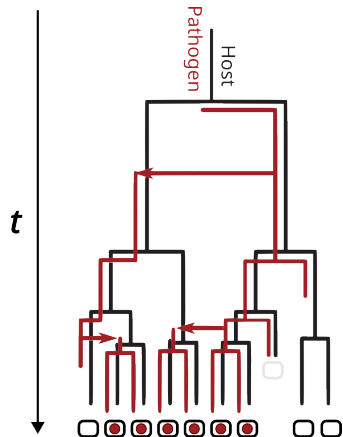
Transmission types

horizontal:

Infection dynamics driven by host-to-host transmission. eg. SI, epidemiological models.

vertical:

Pathogen can be transmitted from infected hosts to their offspring (eg. HIV, Hepatitis B/C, Zika, Toxoplasmosis, ...)



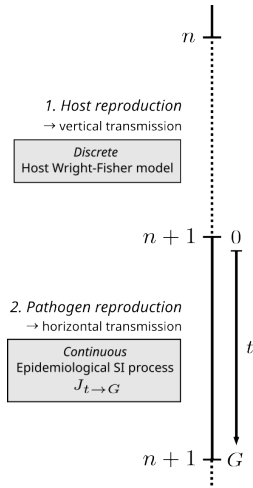
Existing work

- Lipsitch et al. 1995: Epidemiological dynamics of vertical and horizontal transmission (but no coalescence...)
- Welch et al. 2005: Ancestral Infection and Selection Graph models joint host and pathogen genealogies within a Moran model framework.

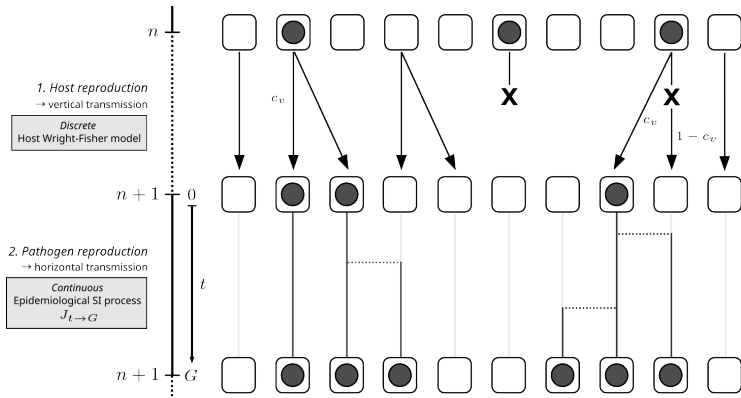
Our approach

- Joint host and pathogen model with intertwined host and pathogen coalescence
- Non-overlapping host generations (Wright-Fisher model)
- Vertical pathogen transmission from one generation to the next
- Epidemiological dynamics (SI model) within one host generation

Semi-discrete nested Host-Pathogen model



Semi-discrete nested Host-Pathogen model



Bibliography

-  Kühnert, Denise, Chieh-Hsi Wu, and Alexei J. Drummond (2011). “Phylogenetic and Epidemic Modeling of Rapidly Evolving Infectious Diseases”. In: *Infection, Genetics and Evolution* 11.8, pp. 1825–1841. DOI: [10.1016/j.meegid.2011.08.005](https://doi.org/10.1016/j.meegid.2011.08.005). URL: <http://dx.doi.org/10.1016/j.meegid.2011.08.005>.
-  Lipsitch, Marc et al. (1995). “The Population Dynamics of Vertically and Horizontally Transmitted Parasites”. In: *Proceedings: Biological Sciences* 260.1359, pp. 321–327. ISSN: 09628452. URL: <http://www.jstor.org/stable/50194> (visited on 09/07/2026).
-  Plessis, Louis du and Tanja Stadler (2015). “Getting To the Root of Epidemic Spread With Phylodynamic Analysis of Genomic Data”. In: *Trends in Microbiology* 23.7, pp. 383–386.